DEL MAR
EST. SEAFOODS 1988

NOTES FROM QINGDAO · 青岛会谈备忘

PLATE FREEZER PROJECT NOTES

平板冻结项目备忘

What we talked about and what we're after, so Wolfking can get started. English first, then Chinese.
我们谈到的内容和我们想要的东西，方便狼王着手。先英文，后中文。

DEL MAR SEAFOODS · 30 OCTOBER 2026 · FOR WOLFKING AND APCCO

ENGLISH

A note on these notes: they were put together with AI from the meeting transcripts and the email chains, and I don't have the refrigeration or regulatory expertise to check every line. If anything here conflicts with what Wolfking or APCCO know, they're right and this is wrong.

WHO'S DOING WHAT

- **Del Mar (Joe)** — the customer. Picks the block size, approves the layout and the final design.
- **Wolfking** — designs and builds four vertical plate freezers with hydraulics and controls, plus the ammonia package to run them. Supplies drawings and the paperwork the permits need, and two technicians to help APCCO install and start up.
- **APCCO** — our refrigeration contractor in California. Checks the design against California rules, stamps and files the permits, installs everything and services it afterwards. They don't build freezers. They'll talk directly to your engineers and ask a lot of questions; that's just the permit process.
- One email thread, all three of us copied.

> **Sean, about the Poland story.** Regulations are on us, not you. Our job (with APCCO) is to work out what's needed here and order the right thing; yours is to build machines that match the spec and work. APCCO will send you the list of certificates and documents we need before anything is built; you tell us which you can provide. Anything missed on the regulatory side after that is our mistake to fix, not yours.

WHAT WE'RE MAKING

- **Whole California squid** in 10 kg blocks, a test for the China market. Squid season starts in May, so we want to be running by May 2027.
- **Ground salmon racks** (frames left after filleting, ground to a paste) for a pet-food customer, year-round. That contract pays for the machines.

45 t	2 × 10 h	4	140
a day (50 short tons)	shifts + 4 h cleaning	vertical plate freezers	blocks of 10 kg per batch, per freezer (the wish)

Our math: 90 min freeze + 30 min unload/reload = a batch every 2 hours, 4 per shift, 8 a day. 45 t ÷ 8 ÷ 4 freezers ≈ 1.4 t, or 140 blocks, per freezer per batch. A bit lower is okay.

Today we blast-freeze squid in cartons on racks, a lot of hand labor at $30 an hour. These four are a test. If it works we add more of the same design, and later automate loading, unloading, bagging and palletizing.

THE BLOCK

- 10 kg, about 12 litres (12,000,000 mm^3); squid is roughly 20% lighter than water.
- **75 mm thick.** Fixed.
- Face around 535 × 300 mm. We're flexible on the exact face: pick what fits your plates and still gives 12 L. It becomes our standard block for every freezer after this, so let's agree it before plates are ordered.
- Two dividers per station, three blocks across. Check that works with your plate height.
- 90 minutes is our guess for 75 mm. Give us the real number for whole squid and for salmon paste.
- **Drainage matters.** Whole squid dumps water and ink when loaded; the station bottom has to let it out so the customer gets squid, not ice. The paste must not leak out. We need a bottom that handles both. Photos or samples on request.

THE FREEZERS

- Four identical vertical plate freezers, indoors, top-loaded. Hydraulic open/close and lift, hot-gas release, blocks out one side onto a table or conveyor.
- Loading manual or semi-automatic to start (conveyor with gates for squid, hose and pump for paste; your ideas welcome). Build in the interface for your automatic loading and unloading later: top clearance, mounting points, spare I/O.
- Blocks go into bags and are stacked on pallets; no boxes. Quote your partner's bagger as an option.
- Aluminum plates. Stainless or aluminum wherever product or wash water touches; nothing galvanized above the plates. Base frame: quote stainless and hot-dip galvanized with the difference. Salt water and chlorine washdown daily.
- Hydraulic unit: 460 V / 60 Hz, out of the cold and wet area, H1 food-grade oil, −35 °C at the cylinders. One per freezer or shared?
- Stainless ammonia-rated hoses, replaceable from outside. No copper or brass near the ammonia.
- E-stops, guarding at the open top, lockable disconnect. APCCO will check against US rules.
- English manuals, drawings, schematics, a two-year spares list, STEP models.

REFRIGERATION

- **Ammonia (R717).** Freon isn't an option in California at this size, even though it works for your Oregon customers; details in the California section. CO_2 would also pass if that's something you do.
- Wolfking supplies the whole package: compressors, condenser, vessels, pumps, valves, controls. APCCO does the site piping and wiring.
- **Containerized engine room, built as skid-mounted portable equipment:** on skids in the yard next to the building, hard-piped to the freezers inside, liftable and movable as one unit (lifting points, one electrical connection, flanged pipe connections). Joe and APCCO pick the spot in early November; send container dimensions and skid weight before then. Joe also asked about a small self-contained unit per freezer: realistic with ammonia? Cost?

- **Keep the ammonia charge as low as you can.** 500 lb (227 kg) is where California's extra paperwork starts. If we can't get under it, just tell us the number.
- APCCO's early estimate (before 140 blocks): about 45 TR at −30 °F (−34 °C) suction, pumped recirculation. Do your own load calc for the final batch size.
- Tell us: system type and recirculation ratio; suction and condensing temperatures; condenser type (evaporative or air-cooled; water and noise both matter); receiver or not; defrost method and what the freezers need for it; which valves come mounted, loose, or from APCCO, with brands.
- ASME U-stamp vessels, B31.5 piping, plates rated for discharge pressure (hot gas hits them). APCCO gives the numbers.

POWER

- 480 V, 3-phase, 60 Hz; motors nameplated 460 V / 60 Hz. No 380 V and no 220 V single-phase in the US.
- Controls on 120 VAC and 24 VDC.
- Panels to UL 508A (UL or ETL listed), NEMA 4X stainless where there's washdown, name-brand UL components (Schneider, ABB, Siemens, Eaton, Allen-Bradley).
- Motors NEMA Premium / IE3 (required for imports, even inside a machine). Size pumps and fans for 60 Hz speed.
- **Send a load list early:** kW, amps and starting current per freezer and per skid. We need it for the power company, which can take longer than building the machines.
- PLC and screens in English. Remote access is fine. Siemens or Allen-Bradley is easiest to service here.

THE SITE

- Watsonville, California, on Monterey Bay, about 150 km south of San Francisco. Mild coastal weather; APCCO will send condenser design temperatures.
- Send weights, footprints, floor loads and lifting points for the freezers and the skid.
- Ship to Oakland or Long Beach, truck to Watsonville. Keep the package within container or road-legal size.

CALIFORNIA THINGS TO KNOW WHILE YOU DESIGN

More detail here because it drives the design. APCCO owns all of it and gives the exact numbers.

REFRIGERANT: WHY NOT FREON

- California (CARB), since 2022: a new system over 50 lb of charge in a new facility needs a refrigerant with GWP under 150. R-404A is about 3,900, R-507 about 4,000, R-448A/R-449A about 1,390, even R-32 is 675. Under 150 leaves ammonia (0), CO_2 (1), or A2L blends like R-454C/R-455A (about 148), which we don't want at this size.

- Federal (EPA): process refrigeration with evaporators between −30 and −50 °C can't use GWP 700 or more from 1 January 2028. Our later freezers would land after that; we won't build on a refrigerant we can't repeat.
- Over 50 lb of any HFC also means California's refrigerant management program: registration, leak inspections, reporting. Ammonia and CO_2 are outside it.
- So: ammonia, as planned. CO_2 also passes.

AMMONIA RULES

- Design to ANSI/IIAR 2-2021; install, start up and inspect to IIAR 4, 5 and 6; plus the California Mechanical Code. APCCO works to these daily.
- On the drawings: high-side design pressure (typically 250 psig / 17.2 bar; plates and hot-gas piping carry it; APCCO confirms); relief valves on every vessel, piped to a safe point outdoors; a labelled emergency shutoff; pipe and valve labels (refrigerant, pressure level, direction); ammonia detectors in the freezer room and at the package, wired to alarm and shut down.
- Ammonia equipment inside a room makes it a "machinery room" (ventilation, detection, door controls, restricted access, no open flames). The outdoor package avoids most of that. Tell us whether your container is built as a machinery room (fans, detectors, door switches) or as open-air equipment.
- Paperwork by charge: any ammonia means a hazardous-materials plan with the county; 500 lb or more adds CalARP (risk management plan, hazard review, inspections); 10,000 lb adds federal PSM/RMP, which we'll never reach. Lowest charge, least paperwork, and we need the number early.
- Leave room for an eyewash and shower at the package.

VESSELS AND PIPING

- Vessels: ASME Section VIII Div. 1, U-stamp, National Board number. Cal/OSHA permits each ammonia vessel and wants weld radiography; APCCO says which and how much.
- Piping: ASME B31.5; welders to ASME IX; material certificates; pressure-test records for anything shipped pre-piped.
- No copper or brass in the ammonia circuit; hose data sheets; plate pressure rating and test pressure on paper.

ELECTRICAL

- California Electrical Code is the NEC plus California additions. Every item listed by a US lab (UL, ETL or CSA); the inspector reads the labels. Unlisted gear means a field evaluation: slow, expensive, sometimes fails.
- Panels to UL 508A with the SCCR on the label; NEMA 4X in washdown, 3R or 4 outdoors; NFPA 79 wiring and colours; English labels and drawings.
- A lockable disconnect within sight of each motor, and a main disconnect per freezer and per skid.
- Motors at DOE efficiency (NEMA Premium / IE3) with the efficiency on the nameplate; VFDs and soft starters listed too.

FOOD SAFETY

- FDA: seafood HACCP for the squid, animal-food rules for the paste. Cleanable, corrosion-resistant contact surfaces; no flaking galvanizing or paint over product; drains and slopes; NSF H1 lubricants where product could be touched.

WORKER SAFETY

- Cal/OSHA: guarded pinch points, E-stops, lockout points for electrical and hydraulic energy, English warning labels. Any loading platform needs guardrails, toe boards and proper stairs; include it in the design.

NOISE

- Limits at the property line, lower at night, and we run 20 hours a day. Give condenser and compressor noise (dB(A) at 10 m) with the proposal.

PERMITS, AND WHO FILES WHAT

- The refrigeration package is skid-mounted portable equipment and is handled as equipment, not construction. Keep it that way: lifting points, one electrical connection, flanged pipe connections, nothing cast into the ground.
- Work on the building (feeder to the skid, freezer room, wall penetrations) goes through normal permits: APCCO, on stamped drawings.
- Unaffected by how the package is mounted: the refrigerant rules (anything in one place over 12 months is stationary to CARB and EPA), the county's hazardous-materials plan and CalARP (quantity on site), Cal/OSHA vessel permits and ASME stamps, electrical listing, food safety, worker safety, noise.
- County (CUPA) and CalARP: Del Mar with APCCO. Cal/OSHA vessel permits: APCCO. PG&E service and food-plant registration: Del Mar. The county and PG&E paperwork can take longer than manufacturing, so drawings first.

> **Paperwork you'll be asked for.** Stampable drawings (layout, P&ID, one-line, schematics); design-pressure and relief-valve sizing calcs; ASME U-1/U-1A reports and National Board numbers per vessel; material certificates and welder qualifications; pressure-test records; UL 508A label and SCCR calc per panel; motor efficiency nameplates; weights and lifting points; ammonia charge calc; English O&M manuals. If you can't provide one, say so now and APCCO will find a way around it.

TIMING

- **Now:** these notes; Joe and APCCO pick the layout; Wolfking starts.
- **ASAP:** preliminary drawings, P&ID, load list, ammonia charge estimate, proposal. Permits start from the preliminary design.
- **Build:** about 100 days after design approval, per your estimate at the show.
- **Running by May 2027,** June at the latest. Joe told the pet-food customer six months.

- **2027–28:** automatic loading and unloading, bagging, palletizing, check-weighing.

WHAT WOULD HELP US MOST RIGHT NOW

1. Preliminary layout of one freezer: size, plate count, block-face proposal, loading height, discharge side, weight.
2. Freeze-time check for 75 mm whole squid and salmon paste.
3. Refrigeration concept: system type, P&ID, equipment list, ammonia charge, container dimensions.
4. Electrical load list and one-line.
5. Vessel list with ASME status, and which certificates you can supply.
6. Proposal: scope split with APCCO, price, delivery, terms, spares, technician terms.
7. STEP models when you have them.
8. A couple of US references we can call (Oregon, LA).
9. A ballpark for phase-2 automation and the bagger, so we design the interfaces now.

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中文

关于这份备忘：它是用 AI 根据会谈录音转写和邮件往来整理的，我没有制冷和法规方面的专业能力来逐条核实。如果这里有什么和狼王或 APCCO 的专业判断不一致，以他们为准，是这里写错了。

谁负责什么

- **Del Mar（Joe）**——客户。定冻块尺寸，批准布置和最终设计。
- **狼王**——设计制造 4 台立式平板冻结机（含液压和控制）和配套氨机组。提供图纸和办许可证所需的资料，派两名技术人员协助 APCCO 安装和开机。
- **APCCO**——我们在加州的制冷承包商。按加州法规审核设计，盖章申报许可证，安装全部设备，之后负责维护。他们不造冻结机。他们会直接和你们的工程师沟通，问很多问题，这是办许可证的正常流程。
- 一个邮件线程，三方都抄送。

> **Sean，关于波兰那件事。**法规是我们的事，不是你们的。搞清楚这边需要什么、下对订单，是我们（和 APCCO）的工作；你们的工作是把机器按规格做好、能用。开工前 APCCO 会把我们需要的证书和文件清单发给你们，你们告诉我们哪些能提供。之后法规方面再有遗漏，是我们的问题，我们自己解决，不找你们。

我们要做什么

- **整条加州鱿鱼**，10 kg 冻块，面向中国市场的试水。鱿鱼季 5 月开始，所以希望 2027 年 5 月开机。
- **三文鱼骨架绞成的鱼糜**（切片后剩下的骨架，绞成糊状），供宠物食品客户，全年生产。这个合同付设备的钱。

45 t	2 × 10 h	4	140
每天（50 美吨）	两班 + 4 小时清洗	台立式平板冻结机	每台每批 10 kg 冻块数（期望值）

我们的算法：冻结 90 分钟 + 卸装料 30 分钟 = 每 2 小时一批，每班 4 批，每天 8 批。45 t ÷ 8 ÷ 4 台 ≈ 每台每批 1.4 t，即 140 块。略少一点也行。

现在我们是鱿鱼装纸箱、上架、进鼓风冻结间，人工太多，这里人工每小时 30 美元。这 4 台是试验。做得通就按同样设计加机器，之后把装料、卸料、装袋、码垛自动化。

冻块

- 10 kg，约 12 升（12,000,000 mm^3）；鱿鱼比水轻 20% 左右。
- **厚度 75 mm。**定死的。
- 平面尺寸约 535 × 300 mm。具体尺寸可以灵活：按你们的冻结板来定，保证 12 L。选定后就是我们今后所有冻结机的标准冻块，所以订板前先定下来。
- 每格两块隔板、并排三块。请核对和板高是否匹配。
- 75 mm 冻 90 分钟是我们的估计。请给出整鱿鱼和三文鱼糜的真实时间。
- **排水很重要。**整鱿鱼装进去会出很多水和墨汁，格底得能放出去，客户拿到的才是鱿鱼而不是冰。鱼糜又不能漏出去。需要两样都兼顾的底部设计。需要的话可寄照片或样品。

冻结机

- 4 台相同的立式平板冻结机，室内，顶部装料。液压开合和顶升，热氨脱模，冻块从一侧出到台面或输送机。
- 装料一开始人工或半自动（鱿鱼用带闸门的输送机，鱼糜用软管加泵，欢迎出主意）。请预留今后加装你们自动装卸料的接口：顶部空间、安装点、备用 I/O。
- 冻块装袋后直接码在托盘上，不装箱。你们合作伙伴的装袋机请作为选项报价。
- 铝合金冻结板。产品和冲洗水碰得到的地方用不锈钢或铝，冻结板以上不要有镀锌件。底座机架：不锈钢和热镀锌分别报价、给出差价。每天都有海水和含氯冲洗。
- 液压站：460 V / 60 Hz，放在低温潮湿区之外，能用 H1 食品级液压油，油缸处耐 −35 °C。每台一套还是共用?
- 不锈钢氨用软管，可从外部更换。氨附近不用铜和黄铜。
- 急停、顶部开口防护、可上锁的电源开关。APCCO 会按美国规定检查。
- 英文手册、图纸、电气图、两年备件清单、STEP 模型。

制冷

- **氨（R717）。**这个规模在加州不能用氟利昂，哪怕你们俄勒冈的客户用得好好的；原因见"加州规定"一节。如果你们做 CO_2，CO_2 也能过关。
- 狼王供整套：压缩机、冷凝器、容器、泵、阀门、控制。现场管道和接线由 APCCO 做。

- **集装箱式机房，做成撬装的可移动设备：**放在厂房旁院子里的撬座上，硬管接到室内冻结机，能整体吊起搬走（吊点、一个电源接口、法兰管道接口）。Joe 和 APCCO 11 月上旬定位置，请在那之前发来集装箱尺寸和撬块重量。Joe 还问过每台冻结机配小型独立机组：用氨现实吗？多少钱？
- **氨充注量尽量做小。**500 磅（227 kg）是加州额外手续的起点。做不到就告诉我们数字。
- APCCO 早先的粗估（140 块之前）：约 45 冷吨，蒸发温度 −30 °F（−34 °C），泵循环。请按最终每批冻块数自己算负荷。
- 请告诉我们：系统形式和循环倍率；蒸发和冷凝温度；冷凝器类型（蒸发式或风冷，用水和噪音都有讲究）；带不带贮液器；融霜方式及冻结机的要求；哪些阀门随机装好、散件、或由 APCCO 供，附品牌。
- 容器 ASME U 钢印，管道 B31.5，冻结板按排气压力设计（热氨会打到板上）。数值 APCCO 给。

电

- 480 V，三相，60 Hz；电机铭牌 460 V / 60 Hz。美国没有 380 V，也没有 220 V 单相。
- 控制用 120 VAC 和 24 VDC。
- 控制柜按 UL 508A（UL 或 ETL 认证），冲洗区用 NEMA 4X 不锈钢柜，元件用带 UL 的大品牌（施耐德、ABB、西门子、伊顿、Allen-Bradley）。
- 电机 NEMA Premium / IE3（进口电机必须，装在整机里也一样）。泵和风机按 60 Hz 转速选。
- **负荷清单请早点给：**每台冻结机、每个撬块的 kW、电流、启动电流。要拿去电力公司申请，可能比造机器还慢。
- PLC 和触摸屏用英文。远程接入没问题。西门子或 Allen-Bradley 在这边最好修。

现场

- 加州沃森维尔（Watsonville），蒙特雷湾，旧金山以南约 150 km。沿海气候温和；冷凝器设计温度 APCCO 会发。
- 请提供冻结机和撬块的重量、占地尺寸、地面荷载和吊点。
- 海运到奥克兰或长滩，卡车到沃森维尔。机组控制在集装箱或公路合法尺寸内。

设计时要知道的加州规定

这一节细一些，因为它直接影响设计。都以 APCCO 为准，具体数值由他们给。

制冷剂：为什么不能用氟利昂

- 加州（CARB）2022 年起：新建设施里充注量超过 50 磅的新系统，制冷剂 GWP 须低于 150。R-404A 约 3,900，R-507 约 4,000，R-448A/R-449A 约 1,390，R-32 也有 675。低于 150 的只有氨（0）、CO_2（1），或 R-454C/R-455A 这类 A2L 混合制冷剂（约 148），这个规模我们不想用。
- 联邦（EPA）：蒸发温度 −30 到 −50 °C 的工艺制冷，2028 年 1 月 1 日起不得用 GWP 700 以上的制冷剂。之后加的冻结机正好落在那之后；我们不会建在一种没法复制的制冷剂上。
- 超过 50 磅氢氟烃还要进加州的制冷剂管理计划：登记、泄漏检查、报告。氨和 CO_2 不在其内。
- 所以：按计划用氨。CO_2 也能过关。

氨的规定

- 设计按 ANSI/IIAR 2-2021；安装、开机、检验按 IIAR 4、5、6；加《加州机械规范》。APCCO 天天按这些干活。
- 落到图纸上：高压侧设计压力（通常 250 psig / 17.2 bar，冻结板和热氨管路也要承受，APCCO 确认）；每个容器有安全阀，排到室外安全点；有标识的紧急切断阀；管道和阀门标识（制冷剂、压力等级、流向）；冻结机车间和机组处有氨探测器，联动报警和停机。
- 氨设备放在房间里，那个房间就成了"机房"（通风、探测、门口控制、限制进入、禁止明火）。室外机组避开大部分。请说明你们的集装箱是按机房做的（风机、探测器、门开关）还是按露天设备。
- 按充注量分的手续：有氨就要向县里报危险品经营计划；500 磅以上加 CalARP（风险管理计划、危害分析、检查）；10,000 磅加联邦 PSM/RMP，我们到不了。充注量最低、手续最少，数字要早给。
- 机组旁留洗眼器和冲淋的位置。

容器和管道

- 容器：ASME 第 VIII 卷第 1 册，U 钢印，National Board 编号。Cal/OSHA 给每个氨容器发许可证并要求焊缝拍片；哪些、多少，APCCO 说。
- 管道：ASME B31.5；焊工按 ASME 第 IX 卷；材质证明；预装管路附压力试验记录。
- 氨回路不用铜和黄铜；软管附数据表；冻结板的压力等级和试验压力要有书面文件。

电气

- 《加州电气规范》就是 NEC 加加州补充。每件设备都要有美国机构认证（UL、ETL 或 CSA），检查员看标签。没认证的要现场评估：慢、贵、可能不过。

- 控制柜按 UL 508A，标签上标 SCCR；冲洗区 NEMA 4X，室外 3R 或 4；接线和线色按 NFPA 79；标识和图纸用英文。
- 每台电机视线内有可上锁的隔离开关，每台冻结机和每个撬块各一个总开关。
- 电机满足美国能源部能效（NEMA Premium / IE3），铭牌标能效；变频器和软启动器也要认证。

食品安全

- FDA：鱿鱼按水产品 HACCP，鱼糜按动物食品规定。接触面易清洗、耐腐蚀；产品上方不能有会掉皮的镀锌或油漆；有排水和坡度；可能碰到产品的地方用 NSF H1 润滑油。

人员安全

- Cal/OSHA：夹点防护、急停、电气和液压能量的上锁点、英文警示。装料平台要有护栏、踢脚板和正规楼梯，请做进设计里。

噪音

- 地界处有限值，夜间更低，我们每天运行 20 小时。请随报价给出冷凝器和压缩机噪音（10 m 处 dB(A)）。

许可证，谁办什么

- 制冷机组是撬装的可移动设备，按设备而不是按建筑处理。请保持这个属性：吊点、一个电源接口、法兰管道接口，不要有浇在地里的东西。
- 厂房上的工作（到撬块的馈线、冻结机车间、穿墙）走正常许可证：APCCO，凭盖章图纸办。
- 不受安装方式影响的：制冷剂规定（在一个地方放超过 12 个月，CARB 和 EPA 都算固定式）、县里的危险品经营计划和 CalARP（看现场氨量）、Cal/OSHA 容器许可证和 ASME 钢印、电气认证、食品安全、人员安全、噪音。
- 县（CUPA）和 CalARP：Del Mar 和 APCCO。Cal/OSHA 容器许可证：APCCO。PG&E 用电和食品厂登记：Del Mar。县里和 PG&E 的手续可能比造机器还慢，所以图纸先行。

> **你们会被要求提供的文件。**可盖章的图纸（布置图、P&ID、单线图、原理图）；设计压力和安全阀选型计算；每个容器的 ASME U-1/U-1A 报告和 National Board 编号；材质证明和焊工资质；压力试验记录；每个控制柜的 UL 508A 标签和 SCCR 计算；电机能效铭牌；重量和吊点；氨充注量计算；英文操作维护手册。哪项提供不了现在就说，APCCO 会想办法。

时间

- **现在：**这份备忘；Joe 和 APCCO 定布置；狼王开始。
- **尽快：**初步图纸、P&ID、负荷清单、氨充注量估算、报价。许可证从初步设计就开始办。
- **制造：**设计批准后约 100 天，按你们展会上的估计。
- **2027 年 5 月开机，**最晚 6 月。Joe 跟宠物食品客户说的是六个月。
- **2027–28 年：**自动装卸料、装袋、码垛、检重。

现在最需要你们提供的

1. 一台冻结机的初步布置图：尺寸、板数、建议的冻块平面尺寸、装料高度、出料侧、重量。
2. 75 mm 整鱿鱼和三文鱼糜的冻结时间核算。
3. 制冷方案：系统形式、P&ID、设备清单、氨充注量、集装箱尺寸。
4. 电气负荷清单和单线图。
5. 容器清单（含 ASME 情况）和你们能提供的证书。
6. 报价：和 APCCO 的范围划分、价格、交期、条款、备件、技术人员条件。
7. STEP 模型，有了就发。
8. 一两个能联系的美国客户（俄勒冈、洛杉矶）。
9. 第二阶段自动化和装袋机的大概预算，好现在把接口设计进去。

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